

2

1



- Define the **global barrier** Λ as the integral of the local barrier

$$\Lambda(\beta) = \int_0^\beta \lambda(\beta') d\beta', \quad \Lambda = \int_0^1 \lambda(\beta) d\beta$$

► Since the $r(\beta, \beta') \approx |\Lambda(\beta') - \Lambda(\beta)|$ with a dense enough annealing schedule

$$\sum_{i=1}^n r(\beta_{i-1}, \beta_i) \approx \Lambda(\beta_n), \quad \sum_{n=1}^N r(\beta_{n-1}, \beta_n) \approx \Lambda$$

- The global barrier controls the asymptotic efficiency of PT

$$\lim_{N \rightarrow \infty} \tau_{\text{NRPT}} = \frac{1}{2 + 2\Lambda}$$

► Δ is an invariant that doesn't depend on the schedule when N is large.

Measures the cumulative rejection along the path

Can be interpreted as the length of the announcing path with respect to the rejection.

Erro is quadratic for linear path but linear for non-linear path

GLOBAL BARRIER

- ▶ Define the **global barrier** Λ as the integral of the local barrier

$$\Lambda(\beta) = \int_0^\beta \lambda(\beta') d\beta', \quad \Lambda = \int_0^1 \lambda(\beta) d\beta$$

- ▶ Measures the cumulative rejection along the path
- ▶ Can be interpreted as the length of the annealing path with respect to the rejection.
- ▶ Since the $r(\beta, \beta') \approx |\Lambda(\beta') - \Lambda(\beta)|$ with a dense enough annealing schedule

$$\sum_{i=1}^n r(\beta_{i-1}, \beta_i) \approx \Lambda(\beta_n), \quad \sum_{n=1}^N r(\beta_{n-1}, \beta_n) \approx \Lambda$$

- ▶ Λ is an invariant that doesn't depend on the schedule when N is large.
- ▶ Error is quadratic for linear path but linear for a non-linear path
- ▶ The global barrier controls the asymptotic efficiency of PT

$$\lim_{N \rightarrow \infty} \tau_{\text{NRPT}} = \frac{1}{2 + 2\Lambda}$$

OPTIMAL SCHEDULE